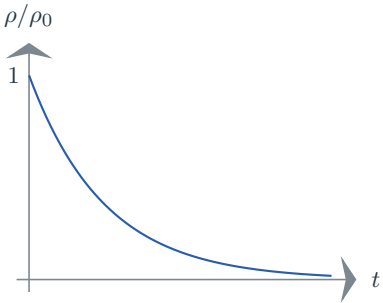
体电荷弛豫由守恒、欧姆关系和高斯定律决定



均匀介质、瞬时欧姆近似有效时；
失去的体电荷可转移到边界

$$\nabla \cdot \mathbf{J} + \partial_t \rho = 0$$
$$\mathbf{J} = \sigma \mathbf{E}, \quad \nabla \cdot \mathbf{E} = \rho/\varepsilon$$

$$\partial_t \rho = -(\sigma/\varepsilon)\rho$$

$$\rho(t) = \rho_0 e^{-t/\tau_c}, \quad \tau_c = \varepsilon/\sigma$$